

Experiment # 1

(ACMP100)

Shashank deep

Optimizing the structure of the molecules and
investigating their Interactions using computational
chemistry (Avogadro Software)

Navigate Tool
Rotate, Translate, Zoom Tool

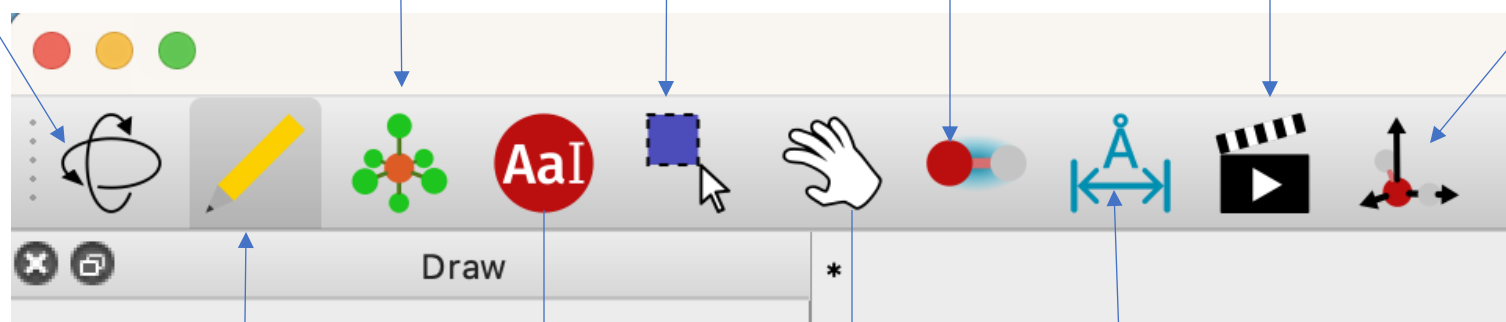
Selection Tool

Animation
Tool

Align Tool

Template Tool

Bond centric
Manipulation Tool

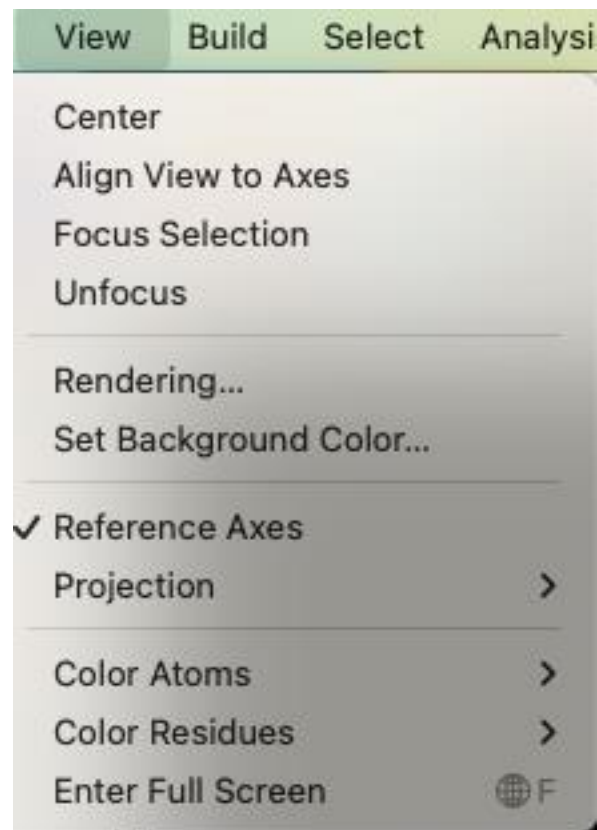
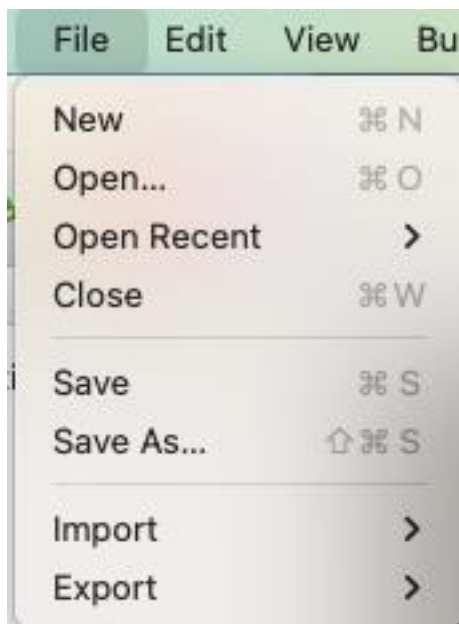
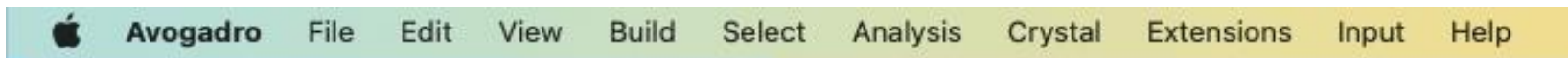


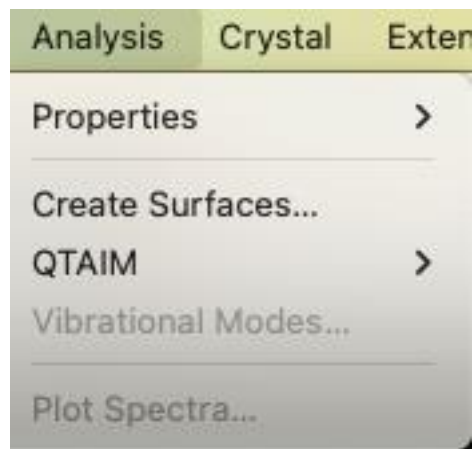
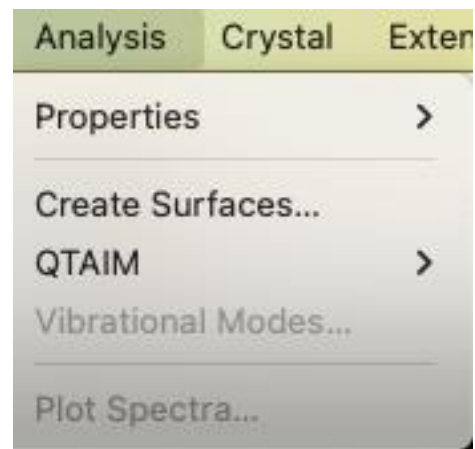
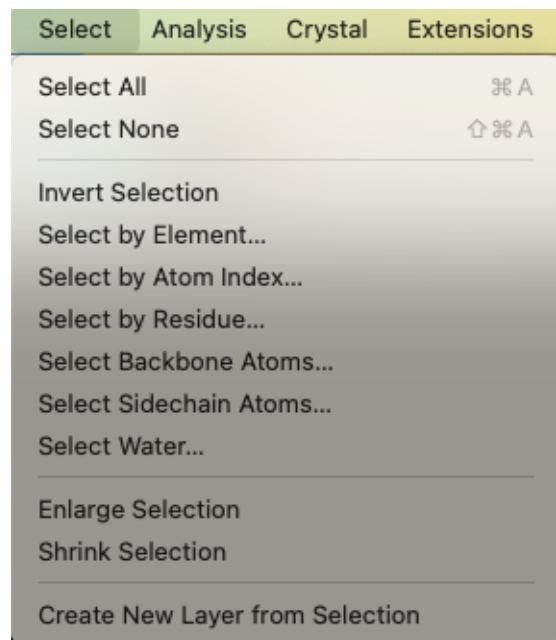
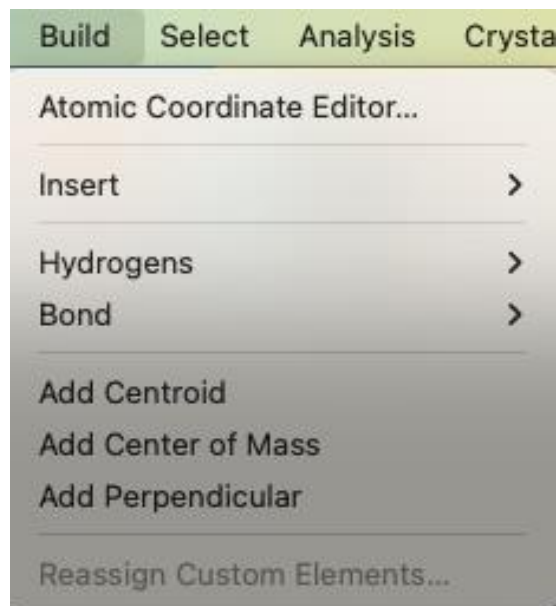
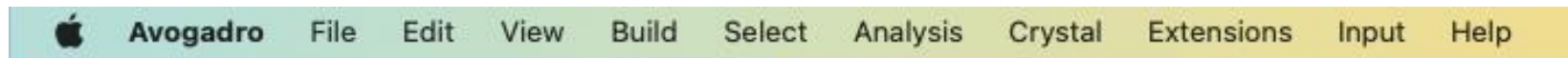
Draw Tool

Atom Label Tool

Measure
Tool

Manipulation Tool





**Avogadro**

File

Edit

View

Build

Select

Analysis

Crystal

Extensions

Input

Help

Crystal Extensions Input Help

Import Crystal from Clipboard...

Add Unit Cell

Edit Unit Cell...

Wrap Atoms to Unit Cell

Rotate to Standard Orientation

Scale Cell Volume...

Build Supercell...

Reduce Cell (Niggli)

Plot Theoretical XRD Pattern...

Space Group



Plot RMSD curve...

Plot Pair Distribution Function...

Extensions Input Help

Optimize Geometry



Calculate



Download Plugins...

Python Settings...

User Interface Language...

Yaehmop



Edit Color Opacity Map...

Periodic Table...

Open Babel



Input Help

APBS



CP2K...

GAMESS...

LAMMPS...

OpenMM Script...

DALTON...

GAMESS-UK...

Gaussian...

MOLPRO...

MOPAC...

NWChem...

ORCA...

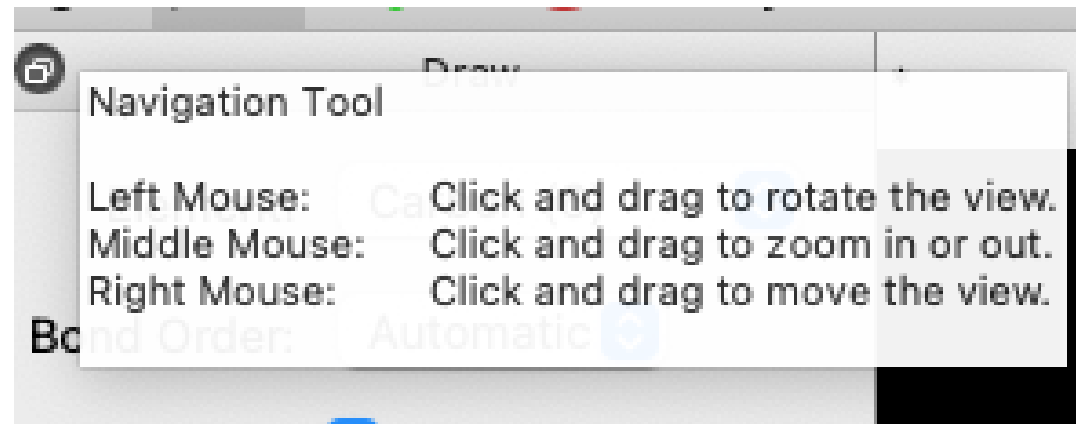
PYSCF...

Psi...

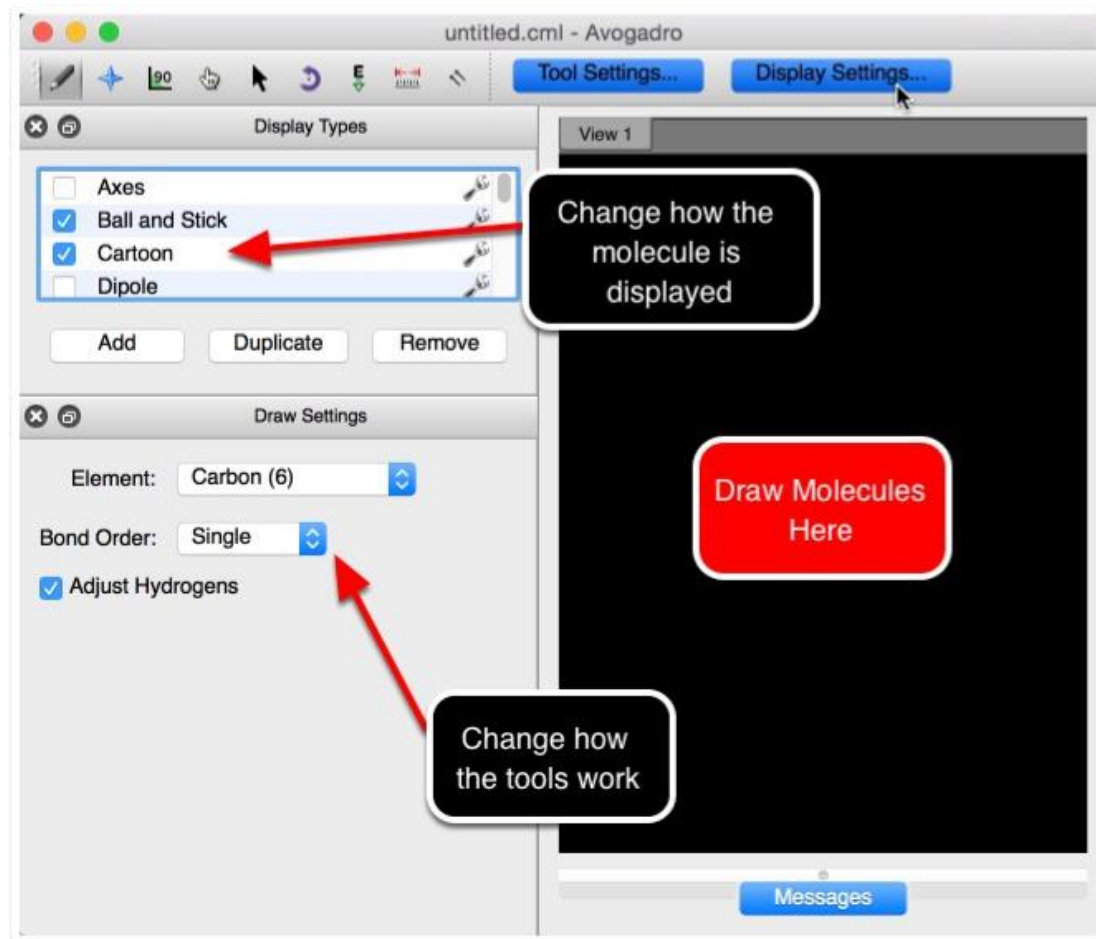
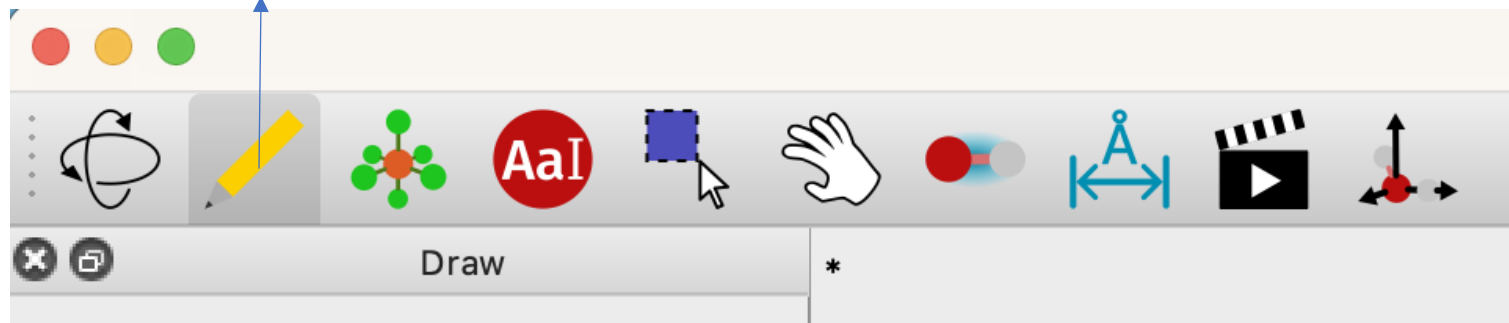
Q-Chem...

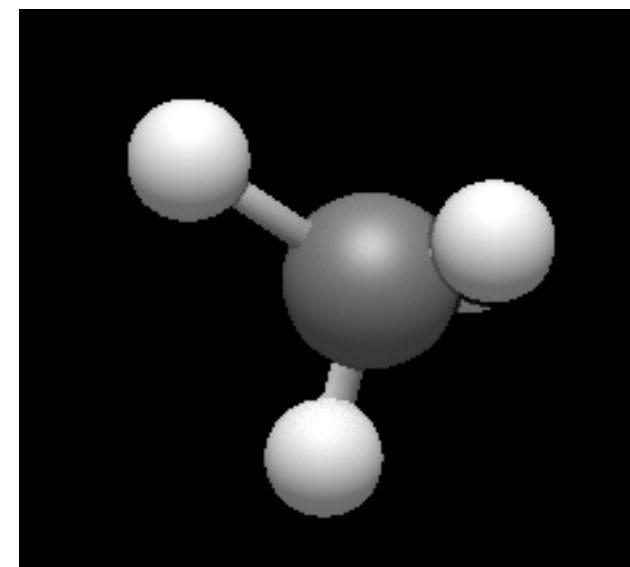
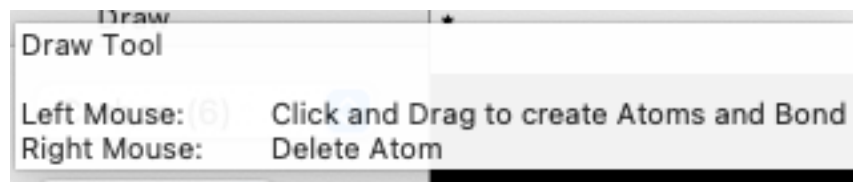
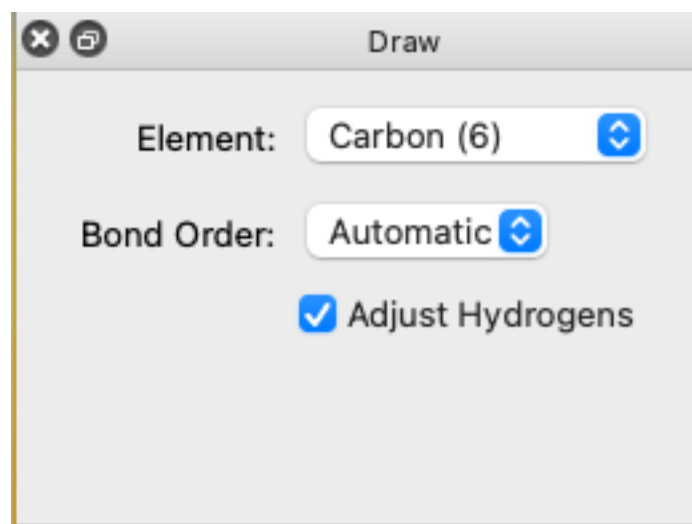
TeraChem...

Navigate Tool
Rotate, Translate, Zoom Tool

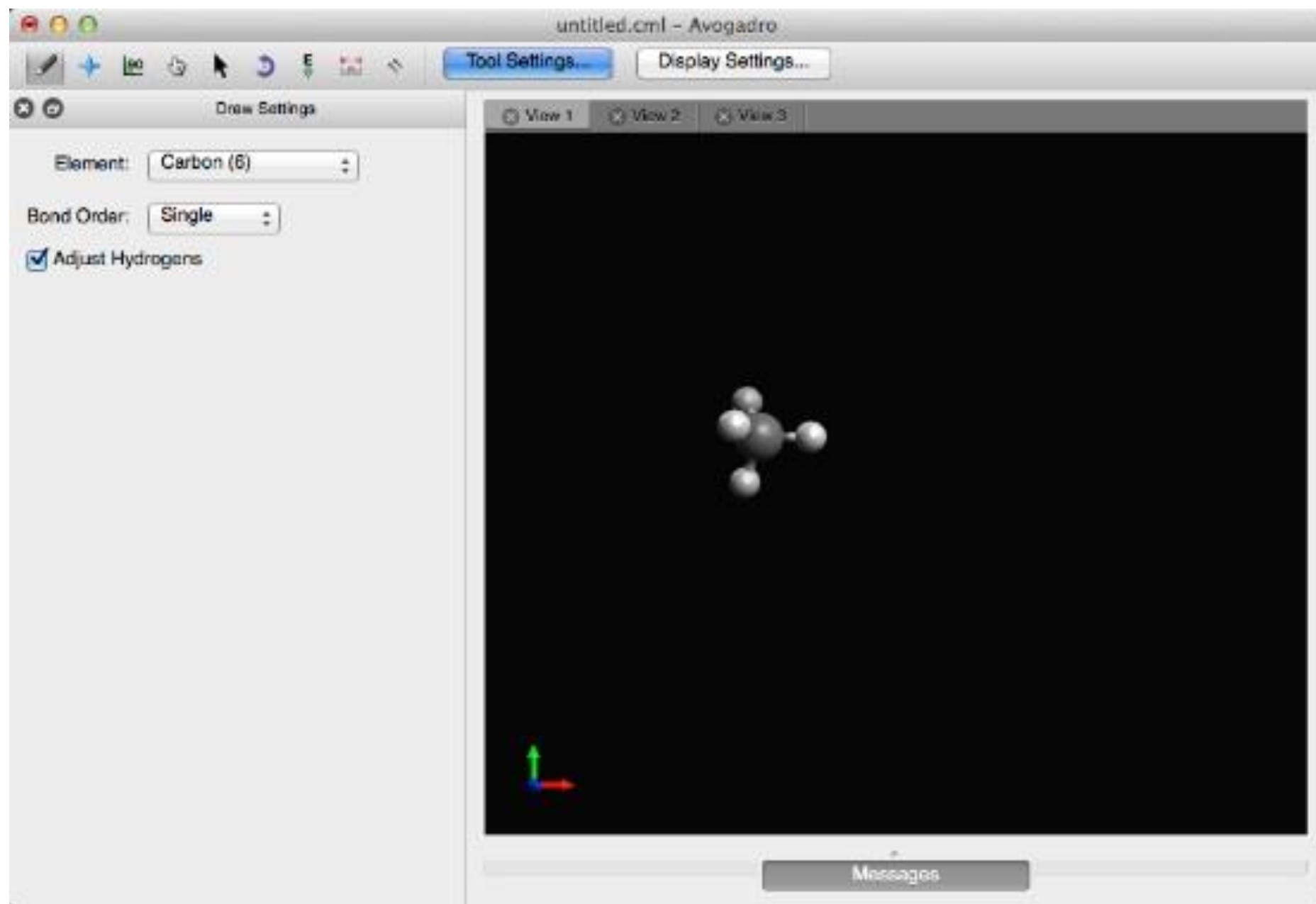


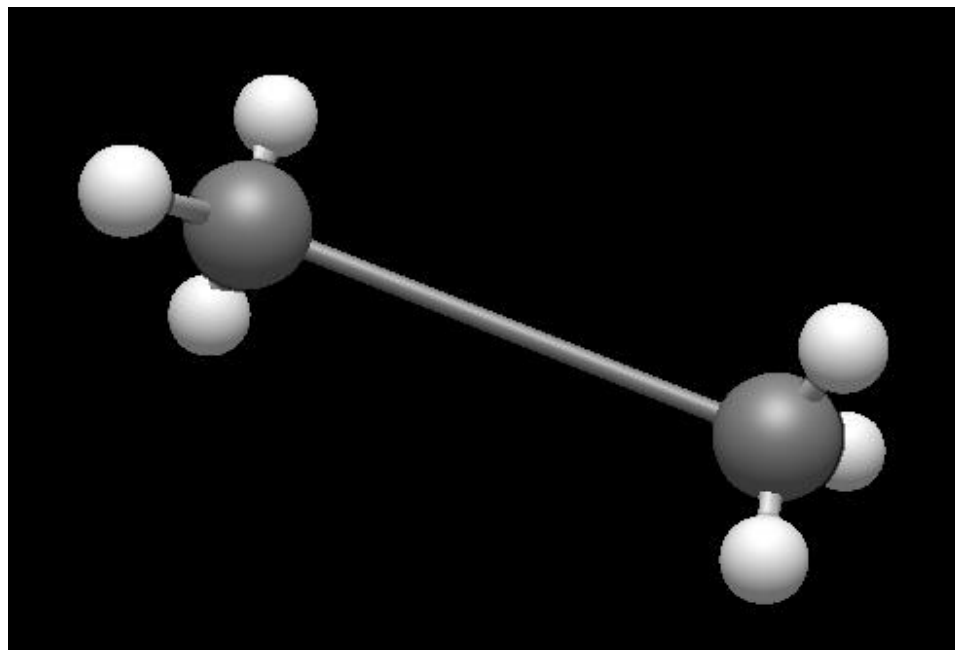
Draw Tool





On the Left Click





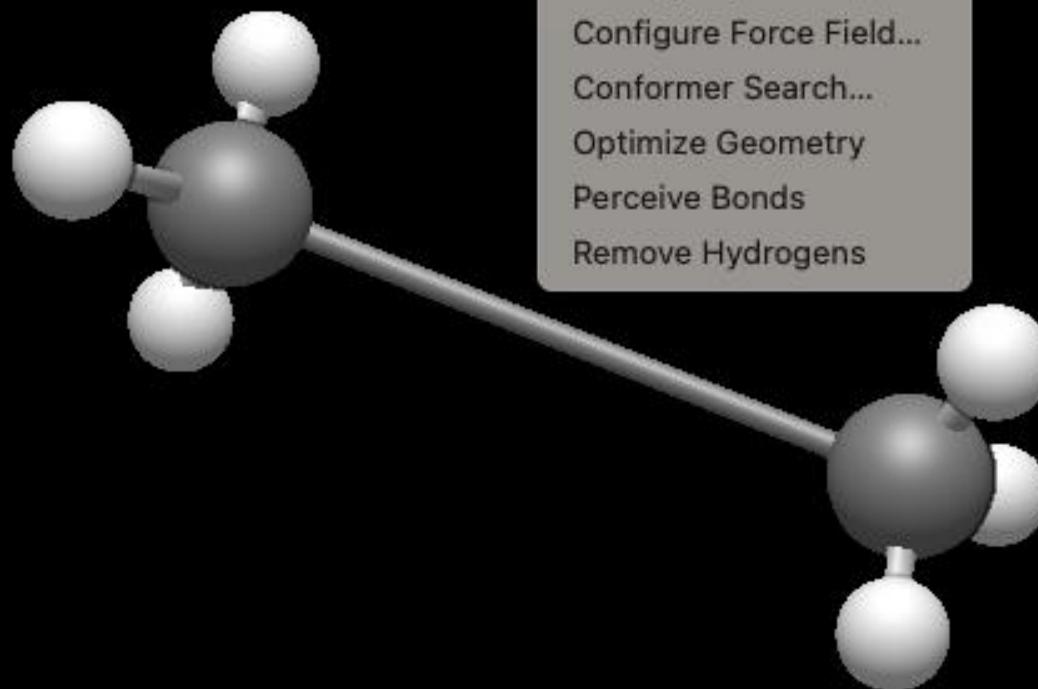
On the Left click and Drag

- Optimize Geometry 
- Calculate 
- Download Plugins...
- Python Settings...
- User Interface Language...
- Yaehmop 
- Edit Color Opacity Map...
- Periodic Table...

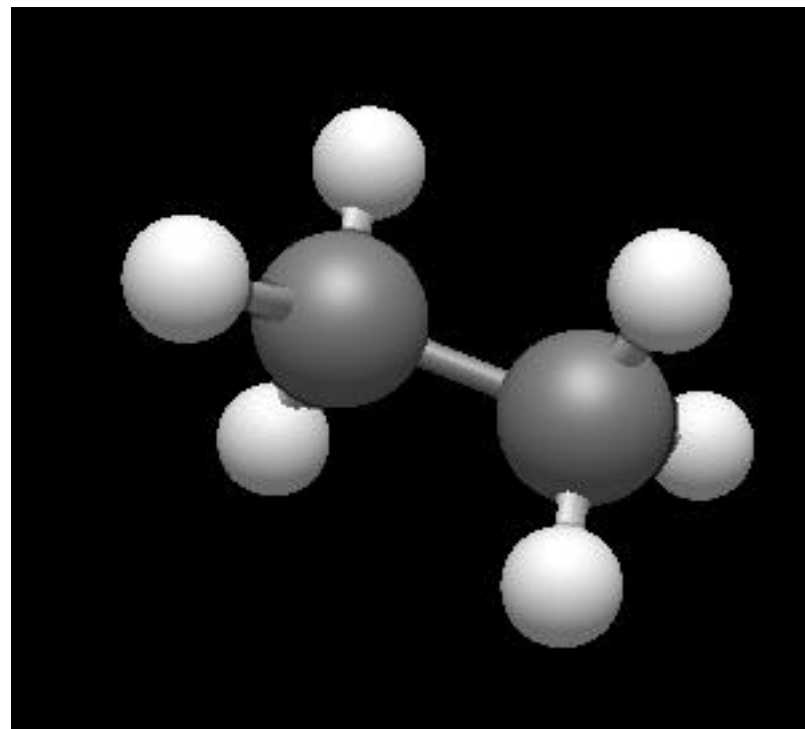
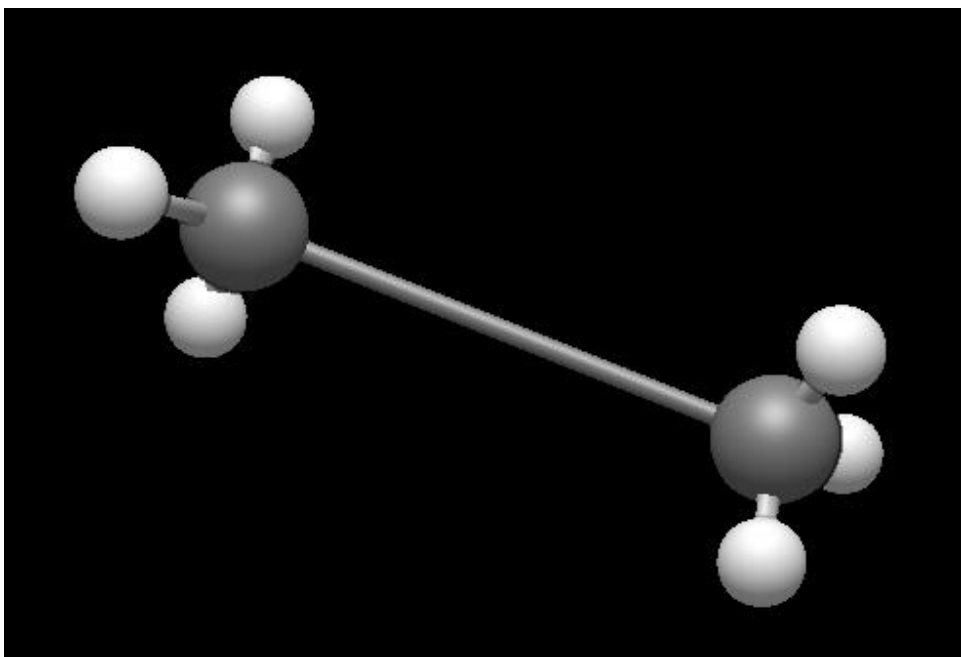
Open Babel 

- Add Hydrogens
- Add Hydrogens for pH...
- Configure Force Field...
- Conformer Search...
- Optimize Geometry
- Perceive Bonds
- Remove Hydrogens

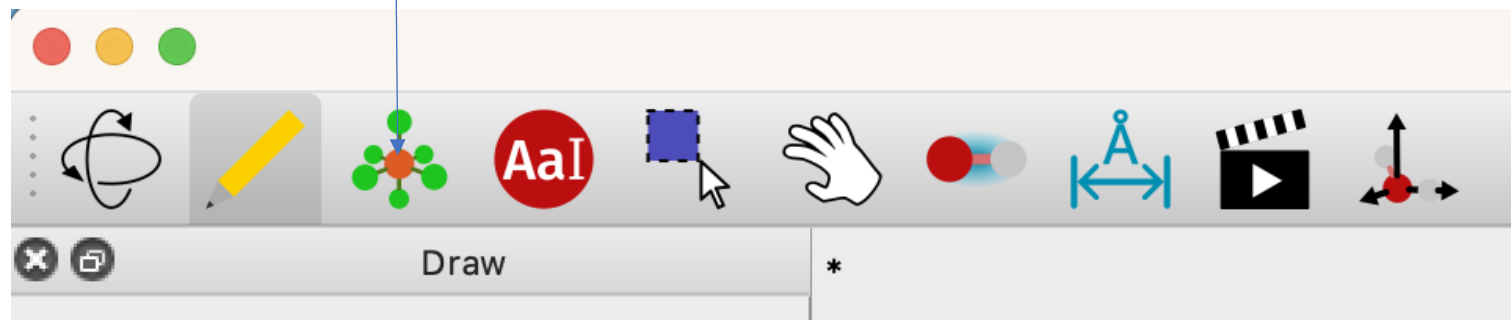
Untitled - Avoga



Click Optimize Geometry



Template Tool



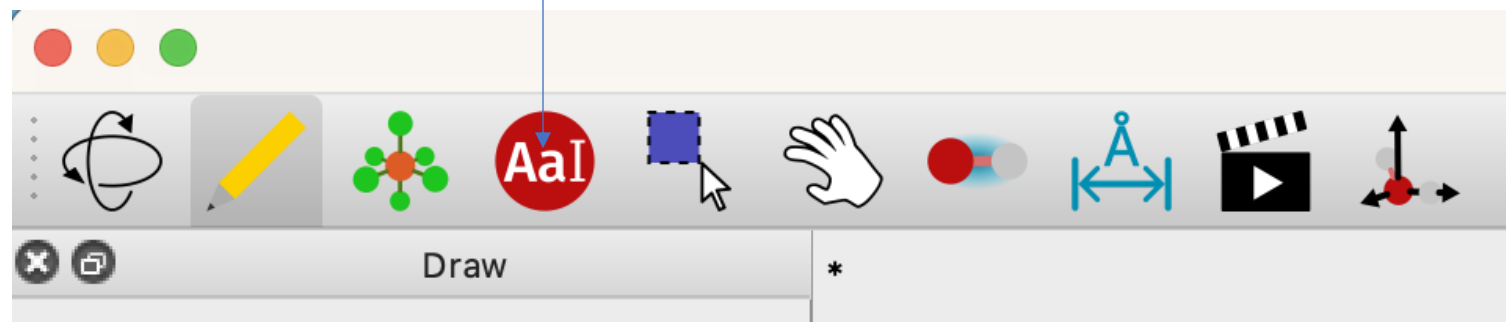
Template Tool

Insert fragments, including metal centers.

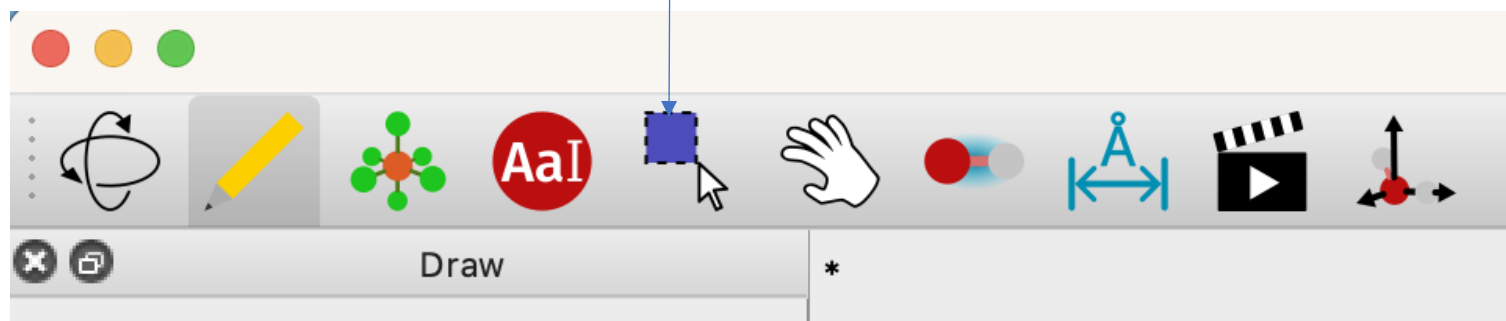
Select an element and coordination geometry, then click to insert a fragment.

Select a ligand or functional group and click on a hydrogen atom to attach it.

Atom Label Tool



Selection Tool



Selection Tool

Left Mouse: Click to pick individual atoms, residues, or fragments

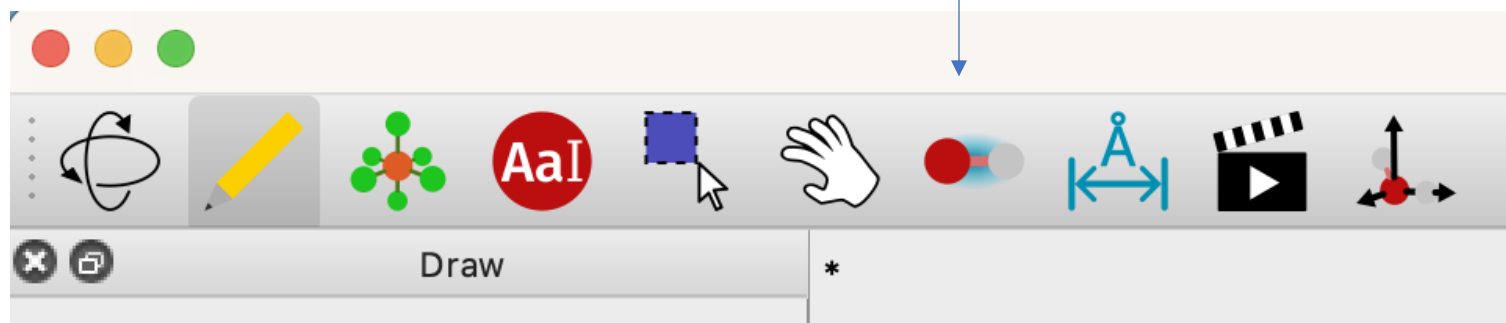
Drag to select a range of atoms

Right Mouse: Click outside the molecule to clear selection

Use Ctrl to toggle the selection and shift to add to the selection.

Double-Click: Select an entire fragment.

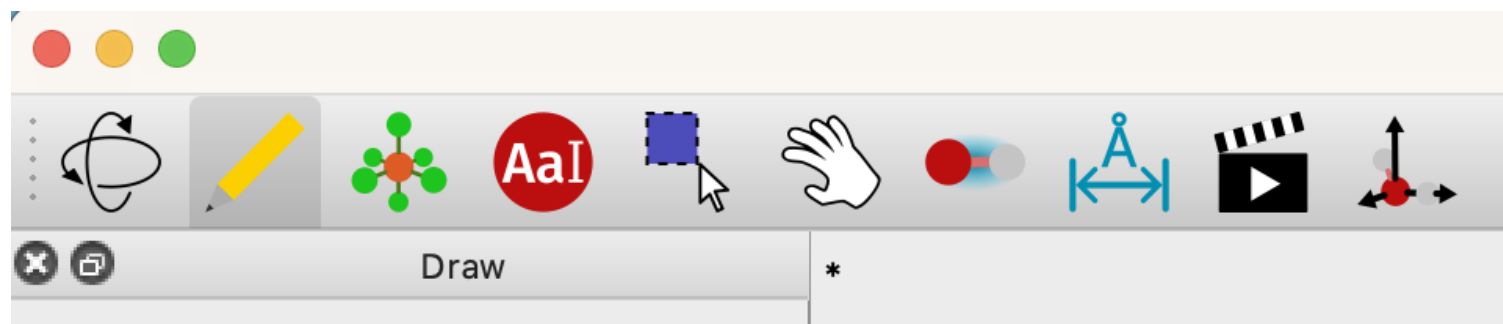
Bond centric
Manipulation Tool



Manipulation Tool

Left Mouse: Click and drag to move atoms

Right Mouse: Click and drag to rotate selected atoms.



Measure Tool

Left Mouse: Select up to four Atoms.

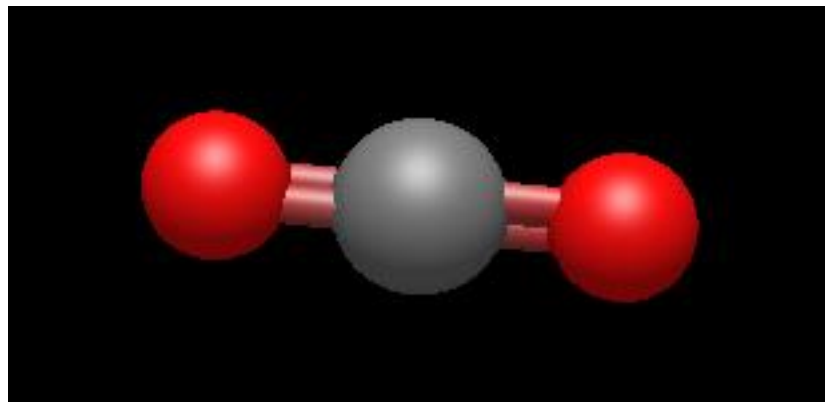
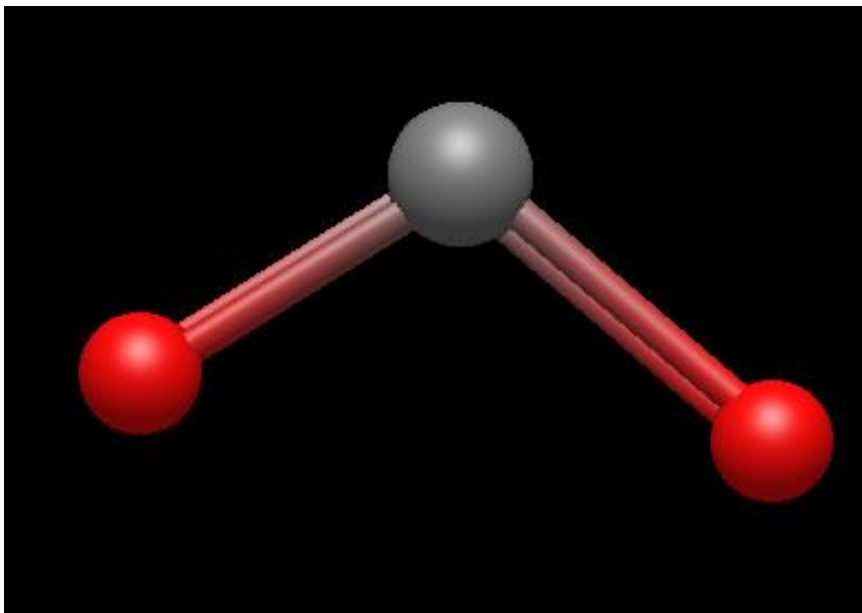
Distances are measured between 1-2 and 2-3

Angle is measured between 1-3 using 2 as the common point

Dihedral is measured between 1-2-3-4

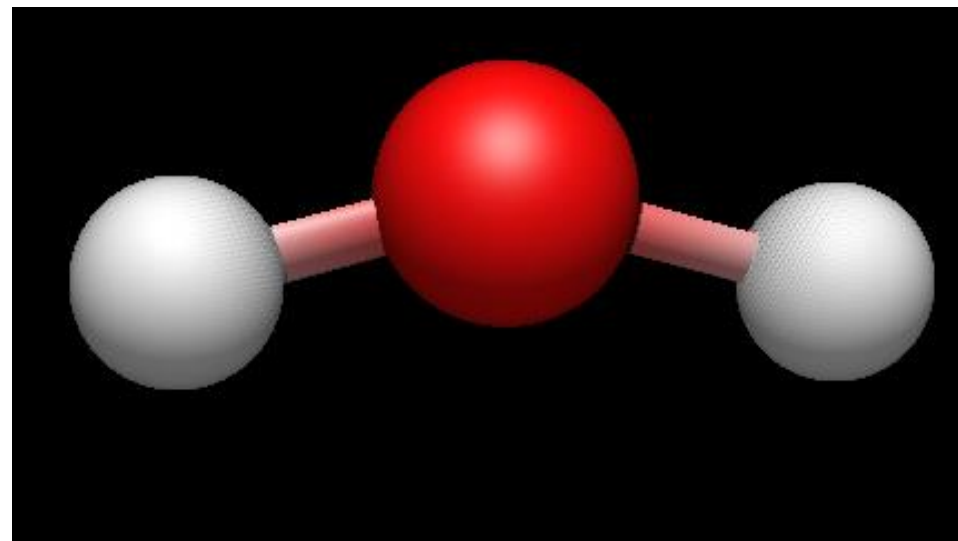
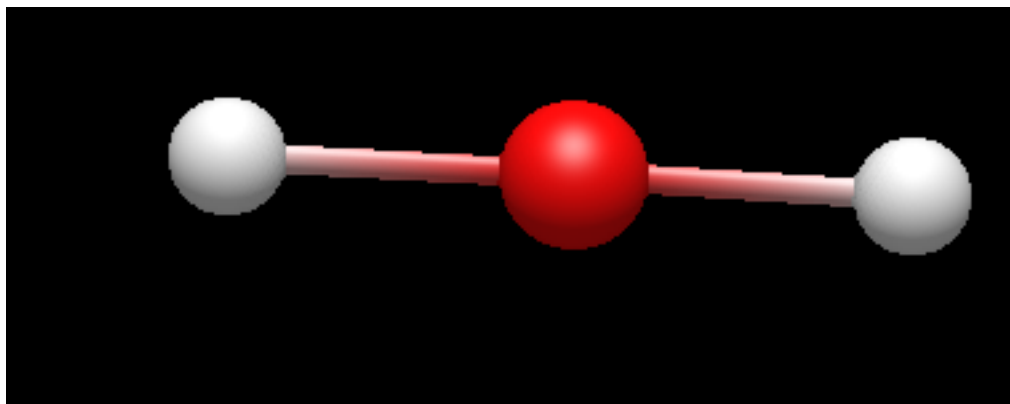
Right Mouse: Reset the measurements.

OPTMIZING THE STRUCTURE OF CO₂



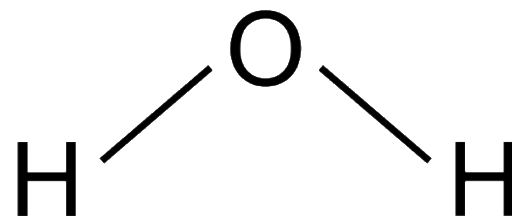
WRITE DOWN BOND LENGTH & BOND ANGLE

OPTMIZING THE STRUCTURE OF H₂O

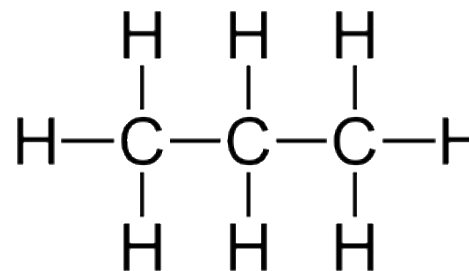


WRITE DOWN BOND LENGTH, BOND ANGLE & ENERGY BEFORE & AFTER OPTIMIZATION

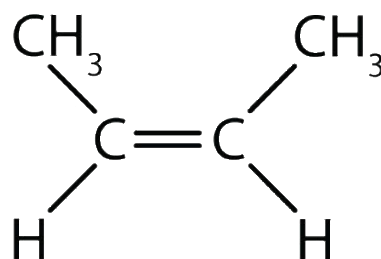
- **Build** and **optimize** the energy of the following **simple** molecules:



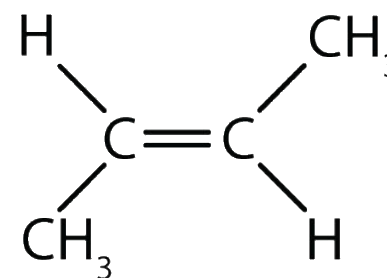
Water



Propane



cis-2-butene



trans-2-butene

WRITE DOWN BOND ANGLE, TORSION ANGLE & ENERGY BEFORE & AFTER OPTIMIZATION

Rotate around Double Bond and optimize

Optimization of Cyclohexane

- Chair or Boat ??

WRITE DOWN ENERGY BEFORE & AFTER OPTIMIZATION

Optimization of 1,2 dichloroethane

- Look at the conformations
- Rotate around Bonds and optimize

Optimization of BCl_3 , NF_3 and PCl_5 and relation to VSEPR theory

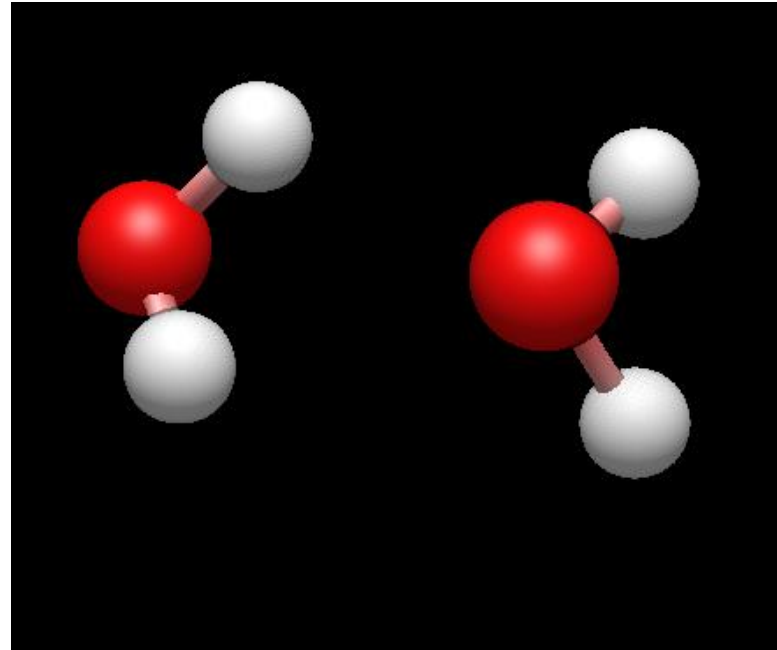
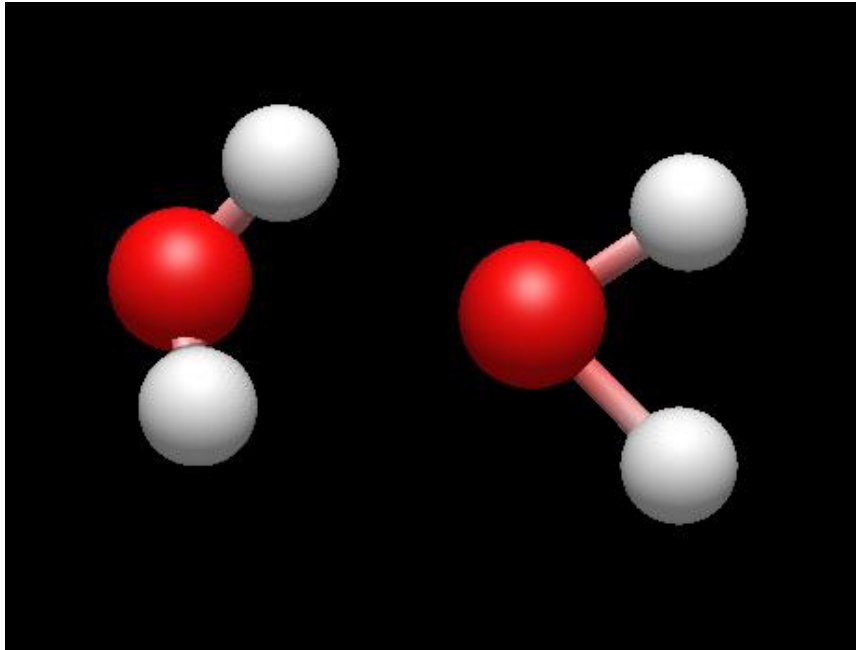
Optimization of P(Ph)_3

Format of Reporting Data

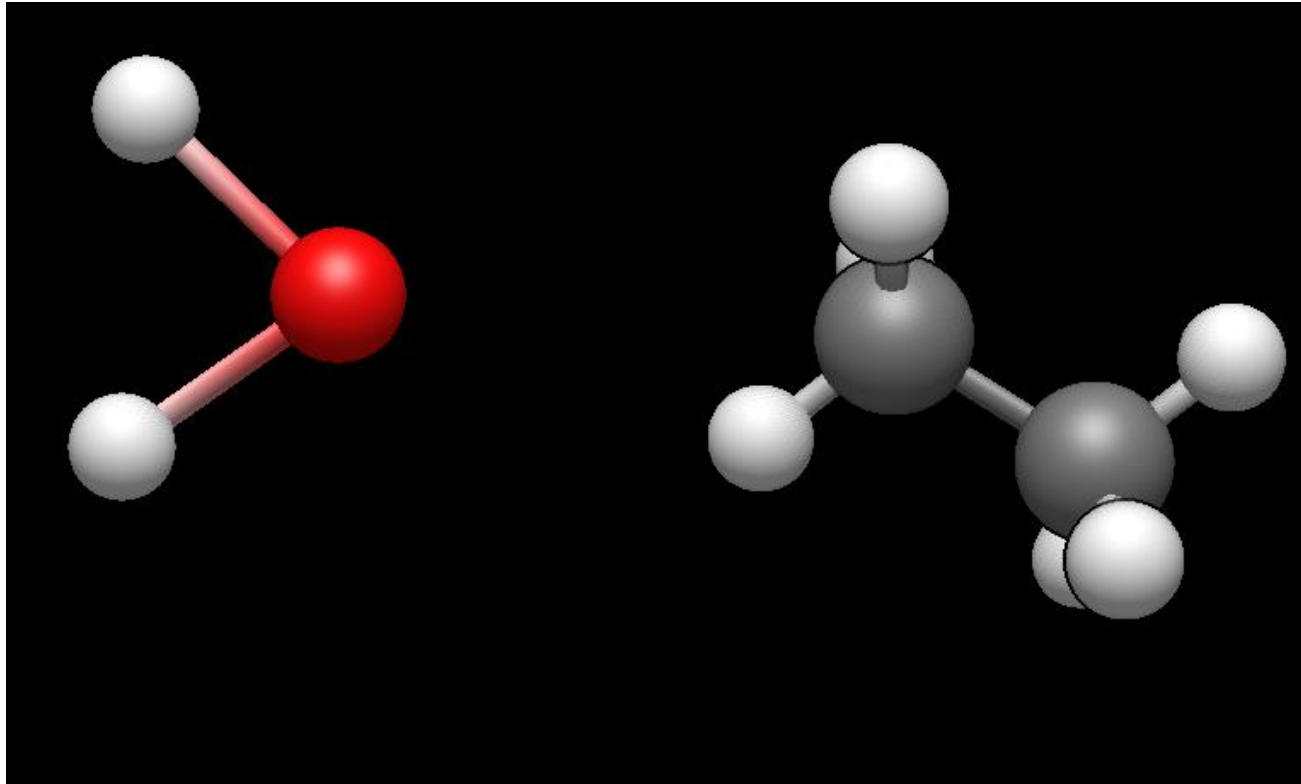
Structure		Energy	Bond length	Bond Angle	Torsion Angle
Molecule Before optimization	Molecule After optimization	Before optimization	Before optimization	Before optimization	Before optimization
		After optimization	After optimization	After optimization	After optimization

Interaction between water and water

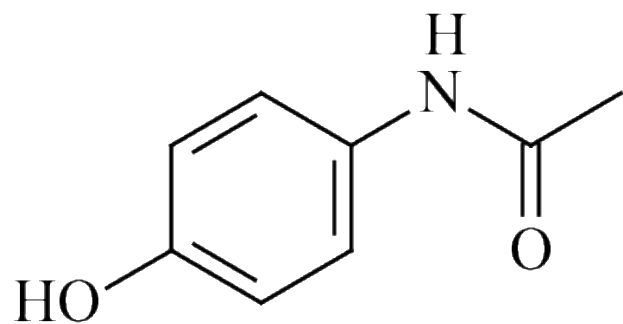
- Note down position of oxygen and hydrogen before and after optimization



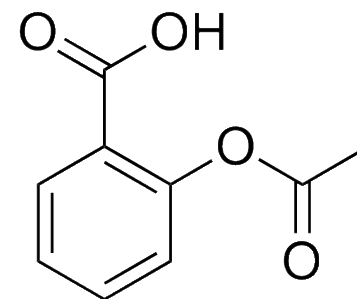
Interaction between water and ethane



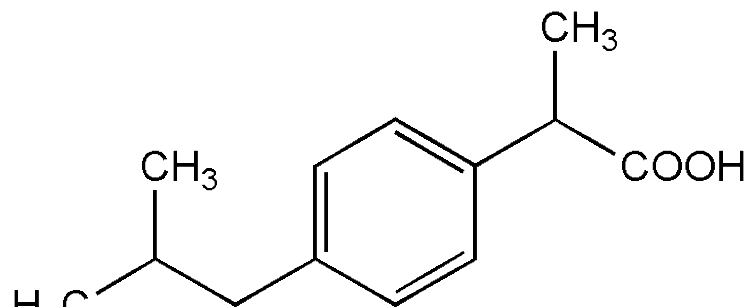
Build and **optimize** the energy of the following drug molecules:



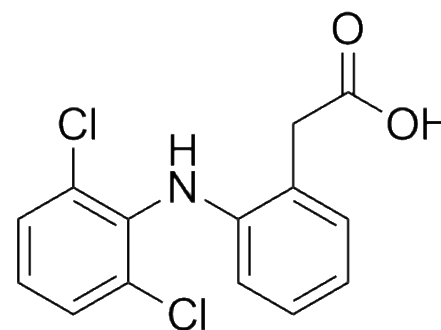
Paracetamol



Aspirin



Ibuprofen



Diclofenac

WRITE DOWN ENERGY BEFORE & AFTER OPTIMIZATION

Build DNA/RNA and peptide